Database Design

Overview

This document serves as the single source of truth for the database design of the Animal Adoption Management System (AAMS). It covers all stages of the design process, from initial entity and relationship identification through to cardinality mapping, table definitions with column names and data types, and the final entity-relationship diagram. All design decisions are grounded in the functional requirements defined for the six user roles: Admin, Shelter Staff, Adopter, Veterinarian, Volunteer, and Donor.

1. Entities & Relationships

This section maps each functional requirement to its corresponding entities and relationships. Relationships are expressed in the format ENTITY_A — verb — ENTITY_B, where each entry represents a structural dependency that will translate into a table, foreign key, or junction table in the final schema. Query-only operations (such as search and filter) are excluded, as they do not produce new schema elements. Relationships are grouped by user role to reflect the access and ownership model of the system.

1.1 Adopter

No	Relationship
01	ADOPTER – has – GOVERNMENT_ID
02	ADOPTER – has – FAVORITES
03	ADOPTER – submits – ADOPTION_APPLICATION
04	ADOPTER – receives – NOTIFICATIONS
05	ADOPTER – cancels – VISITS
06	ADOPTER – takes – ADOPTER_QUIZ
07	ADOPTER – is matched with – PET (through ADOPTER_MATCH)

1.2 Shelter Staff

No	Relationship
01	STAFF – has – GOVERNMENT_ID
02	STAFF – updates – PET
03	STAFF – belongs_to – SHELTER
04	STAFF – manages – APPLICATIONS

No	Relationship
05	STAFF – updates – VOLUNTEERS
06	STAFF – updates – VOLUNTEER_TASKS
07	STAFF – updates – SHELTER_EVENTS
08	STAFF – updates – DONATIONS
09	STAFF – manages – VISITS

1.3 Pet

No	Relationship
01	PET – belongs_to – BREED
02	BREED – belongs_to – SPECIES
03	PET – has – HEALTH_RECORD
04	PET – has – VACCINATION_RECORD
05	PET – has – TRANSFER_HISTORY
06	PET – is matched with – ADOPTER (through ADOPTER_MATCH)
07	PET – has – PET_HEALTH_SUMMARY

1.4 Veterinarian

No	Relationship
01	VET – has – GOVERNMENT_ID
02	VET – manages – APPOINTMENTS
03	VET – updates – VACCINATION_RECORD
04	VET – updates – HEALTH_RECORD
05	APPOINTMENT – records – VACCINATION_RECORD
06	APPOINTMENT – records – HEALTH_RECORD

1.5 Volunteer

No	Relationship
01	VOLUNTEER – HAS – GOVERNMENT_ID
01	VOLUNTEER – submits – VOLUNTEER_APPLICATION
02	VOLUNTEER – has – TASKS
03	VOLUNTEER – manages – SHELTER_EVENTS

Note: Volunteers are assigned to events by staff, never by themselves, through the VOLUNTEER_EVENT rows (staff pass volunteerIDs when creating or editing an event); there

is no self sign-up. Closing a volunteer account, by deactivating or deleting it, removes their VOLUNTEER_EVENT rows for events that have not happened yet. A delete also removes their VOLUNTEER_TASK and VOLUNTEER_EVENT rows, because those foreign keys are RESTRICT and the rows must go before the VOLUNTEER row, while APPOINTMENTS.volunteerID is set to NULL so the appointments they assisted stay.

1.6 DONOR

No	Requirement
01	DONOR – makes – DONATIONS

Note: A donation outlives its donor. DONATION.donorID is nullable with ON DELETE SET NULL (3.28), so a donor deleting their account leaves the shelter's donation history intact; staff see the donor as "Former donor".

1.7 Admin

No	Requirement
01	ADMIN – has – GOVERNMENT_ID
02	ADMIN – creates – SHELTER
03	ADMIN – assigns – STAFF

1.8 Account Security

No	Relationship
01	USERS – has – ACCOUNT_TOKEN

2. Cardinality

This section defines the cardinality of each relationship identified in Section 1. Cardinality describes how many instances of one entity can be associated with instances of another, and is expressed as 1:1, 1:M, or M:M. Many-to-many relationships are resolved into junction tables, which are listed alongside the relationship. These mappings directly inform the placement of foreign keys and the creation of junction tables in the final schema.

No	Relationship	Cardinality	Junction
01	ADOPTER – has – GOVERNMENT_ID	1:1	-
02	ADOPTER – has – FAVORITES	M:M	FAVORITES (adopterID, petID)
03	ADOPTER – submits – ADOPTION_APPLICATION	1:M	-
04	ADOPTER – cancels – VISITS	1:M	-
05	ADOPTER – receives – NOTIFICATIONS	1:M	-

No	Relationship	Cardinality	Junction
06	STAFF – has – GOVERNMENT_ID	1:1	-
07	STAFF – updates – PET	1:M	-
08	STAFF – belongs_to – SHELTER	M:1	-
09	STAFF – manages – ADOPTION_APPLICATIONS	1:M	-
10	STAFF – updates – VOLUNTEER	1:M	-
11	STAFF – assigns – TASKS	1:M	
12	VOLUNTEER – handles – TASKS	M:M	VOLUNTEER_TASKS (taskID, volunteerID)
13	STAFF – updates – EVENTS	1:M	-
14	STAFF – creates – DONOR	1:M	-
15	STAFF – manages – VISITS	1:M	-
16	PET – belongs_to – BREED	M:1	-
17	BREED – belongs_to – SPECIES	M:1	-
18	PET – has – HEALTH_RECORD	1:M	-
19	PET – has – VACCINATION_RECORD	1:M	-
20	PET – has – TRANSFER_HISTORY	1:M	-
21	VET – manages – APPOINTMENTS	1:M	-
22	VET – updates – VACCINATION_RECORD	1:M	-
23	VET – updates – HEALTH_RECORD	1:M	-
24	VOLUNTEER – submits – VOLUNTEER_APPLICATION	1:1	-
25	VOLUNTEER – has – TASKS	1:M	-
26	VOLUNTEER – manages – SHELTER_EVENTS	M:M	VOLUNTEER_EVENT (eventID, volunteerID)
27	DONOR – makes – DONATIONS	1:M	-
28	ADMIN – creates – SHELTER	1:M	-
29	ADMIN – assigns – STAFF	1:M	-
30	APPOINTMENT – records – VACCINATION_RECORD	1:M	-
31	STAFF – assists – APPOINTMENTS	1:M	-
32	VOLUNTEER – assists – APPOINTMENTS	1:M	-
33	APPOINTMENT – records – HEALTH_RECORD	1:M	-

No	Relationship	Cardinality	Junction
34	ADOPTER – takes – ADOPTER_QUIZ	1:1	-
35	ADOPTER – is matched with – PET	M:M	ADOPTER_MATCH (adopterID, petID)
36	PET – has – PET_HEALTH_SUMMARY	1:1	-
37	USERS – has – ACCOUNT_TOKEN	1:M	-

3. Data Tables

This section defines the complete schema for each of the tables in the system. For every table, all columns are listed with their data type, constraints, and a brief description of the field's purpose. Primary keys and foreign keys are explicitly marked. These definitions serve as the direct input for the Prisma schema and database migrations.

3.1 USERS

Central authentication table shared by all six role tables (Admin, Staff, Veterinarian, Adopter, Volunteer, Donor) via userID as a shared primary/foreign key.

Column Name	Data Type	Constraints	Description
userID	INT	PK, AUTO_INCREMENT	Unique identifier for the user, shared across all role tables
userEmail	VARCHAR (45)	NOT NULL, UNIQUE	Login email address
userPassword	VARCHAR (255)	NOT NULL	Bcrypt hashed password
refreshToken	TEXT	NULLABLE	Hashed refresh token for the two-token auth flow
role	VARCHAR (20)	NOT NULL ENUM ('Adopter', 'Volunteer', 'Donor', 'Admin', 'Staff', 'Veterinarian')	Designates which role table this user's profile row lives in
emailVerified	BOOLEAN	DEFAULT FALSE	Whether the user's email address is verified. Account-level, shared by every role. Opening an emailed verification, password-reset, reactivation or email-change link sets it TRUE (see 3.32). Accounts that existed before Sprint 9 were backfilled TRUE (see 3.35)

Column Name	Data Type	Constraints	Description
lastLoginAt	DATETIME	NULLABLE	Most recent successful login (all roles)

Email verification. emailVerified lives on USERS, not on the role tables, so every role shares it (it was moved off the ADOPTER and ADMIN tables). A new account starts FALSE. The migration that created ACCOUNT_TOKEN marked every account that already existed as verified, so the soft gate on applying to adopt and on donating doesn't suddenly block them (see 3.35).

3.2 ADMIN

Stores account and identity information for system administrators who manage shelter branches and staff assignments.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the admin, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
adminName	VARCHAR (45)	NOT NULL	Name of the admin
createdAt	DATETIME	NOT NULL	Account creation timestamp
accountStatus	VARCHAR (45)	DEFAULT 'Pending' ENUM ('Pending', 'Active', 'Deactivated', 'Closed', 'Declined')	Account status (see 3.33)
adminPhone	VARCHAR (20)	NULLABLE	Admin contact number
addressLine1	VARCHAR (100)	DEFAULT ''	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ''	City
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the admin has finished the onboarding wizard

Column Name	Data Type	Constraints	Description
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)
adminDOB	DATE	NULLABLE	Date of birth of admin
adminSex	CHAR (1)	NULLABLE	Gender of admin
statusChangedByID	INT	FK → ADMIN (userID) NULLABLE ON DELETE SET NULL	The admin who last changed this admin's accountStatus (self-referencing)
statusChangedAt	DATETIME	NULLABLE	Timestamp of the last accountStatus change, paired with statusChangedByID

3.3 SHELTER

Stores details for each shelter branch in the multi-shelter network including location, capacity and status.

Column Name	Data Type	Constraints	Description
shelterID	INT	PK, AUTO_INCREMENT	Unique identifier for the shelter
shelterName	VARCHAR (45)	NOT NULL	Name of the shelter
shelterAddress	VARCHAR (45)	NOT NULL	Shelter location
shelterPhone	VARCHAR (20)	NOT NULL	Shelter phone number
shelterEmail	VARCHAR (45)	NOT NULL	Shelter email address
shelterLocation	GEOGRAPHY (POINT)	NULLABLE	Location of the shelter (PostGIS)
shelterZIP	INT	NOT NULL	Zip code of shelter
shelterSize	INT	NOT NULL	Capacity of shelter
shelterStatus	VARCHAR (10)	DEFAULT 'Open' ENUM ('Open', 'Full', 'Closed')	Shelter Status
managerStaffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the shelter manager

3.4 STAFF

Stores account, contact and employment information for shelter staff members.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the staff, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
staffName	VARCHAR (45)	NOT NULL	Full name of the staff
staffPhone	VARCHAR (20)	NULLABLE	Staff contact number
addressLine1	VARCHAR (100)	DEFAULT ''	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ''	City
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the staff has finished the onboarding wizard
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)
shelterID	INT	FK → SHELTER (shelterID) NULLABLE	The shelter branch this staff works at
staffDOB	DATE	NULLABLE	Date of birth of staff
staffSex	CHAR (1)	NULLABLE	Gender of staff
staffDOJ	DATETIME	NULLABLE	Staff joining date
staffDOS	DATETIME	NULLABLE	Staff date of separation
staffDesignation	VARCHAR (45)	NULLABLE ENUM ('Manager', 'Senior', 'Associate')	Staff designation
accountStatus	VARCHAR (45)	DEFAULT 'Pending' ENUM ('Pending', 'Active', 'Deactivated', 'Closed', 'Declined')	Account Status (see 3.33)

3.5 VETERINARIAN

Stores account and contact information for vets who manage animal health appointments and records. A vet onboards (Personal, Address, government ID, Review) before the shelter manager can approve the account. The foreign keys that reference VETERINARIAN (userID) from APPOINTMENTS.vetID, HEALTH_RECORD.vetID and VACCINATION_RECORD.administeredBy are all ON DELETE SET NULL: deleting a vet's account removes the VETERINARIAN row, the USERS row and the vet's GOVERNMENT_ID row (and its Storage file), while the vet's clinical history stays with the pets.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the vet, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
vetName	VARCHAR (45)	NOT NULL	Name of the vet
addressLine1	VARCHAR (100)	DEFAULT ''	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ''	City
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the vet has finished the onboarding wizard; the manager can approve the account only when this is TRUE and the government ID is Verified
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (2–5: Personal, Address, Identity, Review; only ever advances)
vetPhone	VARCHAR (20)	NULLABLE	Vet contact number
vetDOB	DATE	NULLABLE	Date of Birth of vet

Column Name	Data Type	Constraints	Description
vetSex	CHAR (1)	NULLABLE	Gender of vet
createdAt	DATETIME	NOT NULL	Account creation timestamp
shelterID	INT	FK → SHELTER (shelterID) NULLABLE	Unique identifier for the shelter vet associated with
accountStatus	VARCHAR (45)	DEFAULT 'Pending' ENUM ('Pending', 'Active', 'Deactivated', 'Closed', 'Declined')	Account status — Pending until the shelter manager approves (see 3.33)

3.6 SPECIES

Stores the list of animal species supported by the system (e.g. Dog, Cat, Rabbit).

Column Name	Data Type	Constraints	Description
speciesID	INT	PK, AUTO_INCREMENT	Unique identifier for the species
speciesName	VARCHAR (45)	NOT NULL	Name of the species

3.7 BREED

Stores animal breeds and their association to a parent species.

Column Name	Data Type	Constraints	Description
breedID	INT	PK, AUTO_INCREMENT	Unique identifier for the breed
speciesID	INT	FK → SPECIES (speciesID)	Unique identifier for the species the breed belongs to
breedName	VARCHAR (45)	NOT NULL	Name of the breed

3.8 PET

Stores profile, intake, medical and compatibility information for all animals in the shelter network.

Column Name	Data Type	Constraints	Description
petID	INT	PK, AUTO_INCREMENT	Unique identifier for the pet.
petCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "PE000002")

Column Name	Data Type	Constraints	Description
			GENERATED ALWAYS STORED from petID, read-only
petName	VARCHAR (45)	NOT NULL	Name of the pet.
breedID	INT	FK → BREED (breedID)	Pet Breed
petDOB	DATE	NOT NULL	Date of birth of the pet
petWeight	FLOAT	NOT NULL	Pet weight
petHeight	FLOAT	NOT NULL	Pet height
petBGroup	VARCHAR (5)	NOT NULL	Pet blood group
petColor	VARCHAR (45)	NOT NULL	Pet color
petSize	VARCHAR (10)	NULLABLE ENUM ('Small', 'Medium', 'Large')	Size of pet
petPhoto	VARCHAR (300)	NOT NULL	Primary profile photo URL for the Pet
petSex	CHAR (1)	NOT NULL	Pet gender.
petDesc	VARCHAR (500)	NULLABLE	Free-text description of the pet's personality, temperament and behaviour.
microchipID	VARCHAR (45)	NULLABLE	Industry-standard microchip identifier implanted on the pet. Nullable if not yet microchipped.
intakeDate	DATE	NOT NULL	Most recent intake date
intakeType	VARCHAR (45)	NULLABLE ENUM ('stray', 'surrendered', 'transferred')	Most recent intake type
adoptionStatus	VARCHAR (45)	NULLABLE ENUM ('incoming', 'available', 'adopted', 'fostered', 'transferred', 'deceased')	Availability
shelterID	INT	FK → SHELTER (shelterID)	The shelter branch where the pet is currently

Column Name	Data Type	Constraints	Description
			housed.
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff member managing this pet's profile
compatibleWithChildren	BOOLEAN	DEFAULT FALSE	Compatibility with children
compatibleWithPets	BOOLEAN	DEFAULT FALSE	Compatibility with other pets
specialNeeds	BOOLEAN	DEFAULT FALSE	Special needs requirements
featuredFlag	BOOLEAN	DEFAULT FALSE	Marks the pet as featured on the Home page
updatedAt	TIMESTAMP (3)	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Set automatically on every write to the pet. The compatibility matcher compares it with ADOPTER_MATCH.computedAt to tell when cached matches are stale. (Added in migration 20261009120000_pet_updated_at; existing rows started at the migration time.)

3.9 PET_PHOTO

Stores one or more photos per pet as a one-to-many gallery. PET.petPhoto holds the primary/profile photo URL, with additional photos stored here.

Column Name	Data Type	Constraints	Description
photoID	INT	PK, AUTO_INCREMENT	Unique identifier for the photo
petID	INT	FK → PET (petID)	Unique identifier for the pet this photo belongs to
photoURL	VARCHAR (300)	NOT NULL	Photo URL (Supabase Storage)
uploadedAt	DATETIME	NOT NULL DEFAULT now()	Upload timestamp

3.10 Adopter

Stores personal, household, lifestyle and account information for users who register to adopt or foster a pet.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the adopter, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
adopterName	VARCHAR (45)	NOT NULL	Full name of the adopter
adopterDOB	DATE	NULLABLE	Date of birth of adopter
adopterSex	CHAR (1)	NULLABLE	Gender of adopter
createdAt	DATETIME	NOT NULL	Account creation timestamp
adopterRiskFlag	BOOLEAN	DEFAULT FALSE	Risk flag, false by default. Nothing sets it yet (feature A-11 is out of scope); adopters cannot change it
preQualifyFlag	BOOLEAN	DEFAULT FALSE	Flag set if adopter is pre-qualified based on experience and history, used to fast-track applications.
adopterPhone	VARCHAR (20)	NULLABLE	Adopter contact number
housingType	VARCHAR (15)	NULLABLE ENUM ('Apartment', 'House', 'Other')	House type
ownsOrRents	VARCHAR (10)	NULLABLE ENUM ('Owns', 'Rents')	House ownership status
landlordContact	VARCHAR (20)	NULLABLE	Landlord contact
householdSize	INT	NULLABLE	Household size
numChildren	INT	NULLABLE	Number of children at home
employmentStatus	VARCHAR (15)	NULLABLE ENUM ('Unemployed', 'Student', 'Self_employed', 'Employed')	Employment status of the adopter
activityLevel	VARCHAR (10)	NULLABLE ENUM ('Sedentary', 'Medium', 'Active')	Level of activity in lifestyle of adopter

Column Name	Data Type	Constraints	Description
yardAvailable	BOOLEAN	DEFAULT FALSE	Yard availability
petExperience	VARCHAR (10)	NULLABLE ENUM ('No', 'Little', 'Some', 'Very')	Amount of experience with pets
currentPets	INT	DEFAULT 0	Number of pets currently owned by the adopter at time of registration.
preferredBreedID	INT	FK → BREED (breedID) NULLABLE	Preferred breed for adoption
preferredAgeRange	VARCHAR (15)	NULLABLE ENUM ('Young', 'Adult', 'Old')	Preferred pet age for adoption
preferredSize	VARCHAR (10)	NULLABLE ENUM ('Small', 'Medium', 'Large')	Preferred pet size for adoption
openToSpecialNeeds	BOOLEAN	DEFAULT FALSE	Open to adopting special needs pets
adopterType	VARCHAR (10)	NULLABLE ENUM ('Fosterer', 'Owner')	Type of adopter
stripeCustomerID	VARCHAR (255)	NULLABLE	Stripe payment ID
accountStatus	VARCHAR (45)	DEFAULT 'Active' ENUM ('Active', 'Banned', 'Deactivated', 'Closed')	Account status (see 3.33)
addressLine1	VARCHAR (100)	DEFAULT ''	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ''	City
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the adopter has finished the onboarding wizard
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)

3.11 FAVORITES

Junction table linking adopters to pets they have saved to their favorites list.

Column Name	Data Type	Constraints	Description
adopterID	INT	PK, FK → ADOPTER (userID)	Unique identifier for the adopter
petID	INT	PK, FK → PET (petID)	Unique identifier for the pet

3.12 ADOPTION_APPLICATION

Stores adoption applications submitted by adopters for specific pets, including status and assigned staff.

Column Name	Data Type	Constraints	Description
applicationID	INT	PK, AUTO_INCREMENT	Unique identifier for the application.
petID	INT	FK → PET (petID)	Unique identifier for the pet
adopterID	INT	FK → ADOPTER (userID)	Unique identifier for the adopter
createdAt	DATETIME	NOT NULL	Application creation timestamp
staffID	INT	FK → STAFF (userID) NULLABLE	Staff ID of Staff handling the application
shelterID	INT	FK → SHELTER (shelterID)	Shelter ID of Shelter application is associated with.
applicationStatus	VARCHAR (20)	DEFAULT 'Pending' ENUM ('Pending', 'Accepted', 'Rejected', 'Withdrawn')	Application status
applicationCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "APP-00048") — Postgres GENERATED ALWAYS STORED column computed from applicationID, read-only
applicationType	VARCHAR (10)	DEFAULT 'Adopt', ENUM('Adopt','Foster')	Whether this is an adopt or foster application
shelterMessage	VARCHAR (500)	NULLABLE	Adopter's message to the shelter, submitted with the application
staffRemark	VARCHAR (300)	NULLABLE	Staff-authored closing note shown to the adopter once resolved
stripeCheckout	VARCHAR (255)	NOT NULL	The Stripe Checkout Session that paid the application fee

Column Name	Data Type	Constraints	Description
SessionID			
stripePaymentIntentID	VARCHAR (255)	NULLABLE	The associated Stripe PaymentIntent
paymentStatus	VARCHAR (10)	DEFAULT 'Paid', ENUM('Paid')	Payment status of the processing fee
amountPaid	DECIMAL (6,2)	DEFAULT 15.00	Application processing fee actually charged

3.13 VISITS

Stores shelter visit and virtual meet-and-greet appointments scheduled by adopters.

Column Name	Data Type	Constraints	Description
visitID	INT	PK, AUTO_INCREMENT	Unique identifier for the visit
adopterID	INT	FK → ADOPTER (userID)	Unique identifier for the adopter
petID	INT	FK → PET (petID) NULLABLE	Unique identifier for pet associated with the visit. Can be NULL
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for Staff assisting the visit. Can be NULL
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for shelter associated with the visit.
visitTime	DATETIME	NOT NULL	Visit date and time
remarks	VARCHAR (300)	NULLABLE	Any remarks related to the visit
visitStatus	VARCHAR (45)	NULLABLE ENUM ('Confirmed', 'Cancelled', 'Completed')	Visit status — null until a staff member confirms it

3.14 NOTIFICATIONS

Stores system-generated automated reminders sent to users for vaccinations, appointments and tasks.

Column Name	Data Type	Constraints	Description
notificationID	INT	PK,	Unique identifier for the

Column Name	Data Type	Constraints	Description
		AUTO_INCREMENT	notification
recipientID	INT	NOT NULL	Unique identifier for the recipient
notificationType	VARCHAR (20)	ENUM ('vaccination', 'appointment', 'task')	Type of notification
notificationStatus	VARCHAR (10)	ENUM ('Pending', 'Sent', 'Failed')	Status of notification delivery
recipientType	VARCHAR (20)	ENUM ('adopter', 'staff', 'volunteer', ‘veterinarian’)	Recipient user type
sendTime	DATETIME	NOT NULL	Send date and time
textBody	VARCHAR (300)	NOT NULL	Notification text content

3.15 HEALTH_RECORD

Stores medical history entries for each pet including diagnosis, treatment and vet notes. A note is written either while a vet completes an appointment, in which case appointmentID links it to that appointment, or as a standalone note, in which case appointmentID is null. Only the vet who wrote a record can edit it.

Column Name	Data Type	Constraints	Description
recordID	INT	PK, AUTO_INCREMENT	Unique identifier for the record
petID	INT	FK → PET (petID)	Unique identifier for the pet the record belongs to
createdAt	DATETIME	NOT NULL	Creation date
lastUpdated	DATETIME	NOT NULL	Last update date
vetID	INT	FK → VETERINARIAN (userID) NULLABLE ON DELETE SET NULL	Vet who wrote the record; null when that vet's account has been deleted (displayed as "Former vet")
appointmentID	INT	FK → APPOINTMENTS (appointmentID) NULLABLE ON DELETE SET NULL INDEX HealthRecord_appointmentID_idx	Appointment the note was written at while completing it; null for standalone notes. Mirrors VACCINATION_RECORD.appointmentID
recordDesc	VARCHAR (500)	NOT NULL	Description of the medical visit, diagnosis, treatment, or special needs noted by the

Column Name	Data Type	Constraints	Description
			vet. Only the vet who wrote a record can edit it.

3.16 VACCINE

Network-wide vaccine catalogue shared by every shelter; the table has no shelterID. Vets and Admins add and edit entries. A vaccine is identified by vaccineName plus manufacturer, compared case-insensitively; this uniqueness is enforced by a service-level check and no unique index backs it. VACCINATION_RECORD rows reference vaccineID, so an edit to an entry appears on every pet's passport.

Column Name	Data Type	Constraints	Description
vaccineID	INT	PK, AUTO_INCREMENT	Unique identifier for the vaccine
vaccineName	VARCHAR (45)	NOT NULL	Name of the vaccine
manufacturer	VARCHAR (45)	NULLABLE	Manufacturer name; blank values are stored as null
vaccineDesc	VARCHAR (500)	NULLABLE	Description of the vaccine; blank values are stored as null

3.17 VACCINATION_RECORD

Stores vaccination history for each pet including vaccine details, dates and administering vet. Doses are recorded at an appointment: petID, administeredBy, administeredAt (the appointment's shelter) and appointmentID are set server-side.

Column Name	Data Type	Constraints	Description
recordID	INT	PK, AUTO_INCREMENT	Unique identifier for the record
petID	INT	FK → PET (petID)	Unique identifier for the pet the record belongs to
vaccineID	INT	FK → VACCINE (vaccineID)	Unique identifier for the vaccine administered
administeredDate	DATETIME	NOT NULL	Administered date
dueDate	DATETIME	NULLABLE	Due date of the next dose. Null means the vet planned no further dose (senior or immunocompromised pet, titer shows immunity, discontinued non-core vaccine, history carried in at intake); such a dose is never overdue and the passport

Column Name	Data Type	Constraints	Description
			shows "No Further Dose"
administeredBy	INT	FK → VETERINARIAN (userID) NULLABLE ON DELETE SET NULL	Vet who administered the vaccine; null when that vet's account has been deleted (displayed as "Former vet")
administeredAt	INT	FK → SHELTER (shelterID) NULLABLE	Shelter where the vaccine was administered (the appointment's shelter)
appointmentID	INT	FK → APPOINTMENTS (appointmentID) NULLABLE ON DELETE SET NULL	The appointment the dose was given at; null for doses recorded outside an appointment.

Overdue logic. For each active pet (adoptionStatus not adopted or deceased) and vaccine, only the latest dose counts; the pet is overdue for that vaccine when that dose's dueDate has passed. A dose with a null dueDate is never overdue. Passport statuses are Overdue (dueDate passed), Due Soon (dueDate within 30 days), Up to Date, and No Further Dose (dueDate null).

3.18 TRANSFER_HISTORY

Stores the history of inter-shelter pet transfers including the shelters and staff involved.

Column Name	Data Type	Constraints	Description
recordID	INT	PK, AUTO_INCREMENT	Unique identifier for the record
petID	INT	FK → PET (petID)	Unique identifier for the pet the record belongs to
transferStatus	VARCHAR (20)	DEFAULT 'In_Progress' ENUM ('In_Progress', 'Completed', 'Rejected', 'Cancelled')	Transfer Status
transferDate	DATETIME	NOT NULL	Date of transfer
fromShelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the current shelter the pet is at
toShelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the new shelter the pet moving to
fromShelterStaff	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff handling transfer at current shelter
toShelterStaff	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff handling transfer at new shelter
transferReason	VARCHAR (300)	NOT NULL	Reason for transfer

Health passport (universal). HEALTH_RECORD, VACCINATION_RECORD and TRANSFER_HISTORY are read by petID only and are never filtered by shelter, so a pet's full history, including entries made at shelters it has since left, travels with it across inter-shelter transfers.

3.19 APPOINTMENTS

Stores veterinary health appointments for pets including the attending vet and associated shelter.

Column Name	Data Type	Constraints	Description
appointmentID	INT	PK, AUTO_INCREMENT	Unique identifier for the appointmentID
appointmentCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "APT-00123") GENERATED ALWAYS STORED from appointmentID, read-only
petID	INT	FK → PET (petID)	Unique identifier for the pet associated with the appointment
vetID	INT	FK → VETERINARIAN (userID) NULLABLE ON DELETE SET NULL	Vet attending the appointment. The API always assigns a vet to a new appointment; the column is nullable so that a vet deleting their account keeps the pets' appointment history (a null vet displays as "Former vet")
staffID	INT	FK → STAFF (userID) NULLABLE	Staff member assigned
volunteerID	INT	FK → VOLUNTEER (userID) NULLABLE	Volunteer assigned
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the shelter the appointment is at
appointmentDate	DATETIME	NOT NULL	Appointment date and time
appointmentReason	VARCHAR (300)	NOT NULL	Reason for appointment
appointmentStatus	VARCHAR (10)	DEFAULT 'Scheduled' ENUM AppointmentStatus ('Scheduled', 'Completed', 'Cancelled')	Stored appointment status. Completed is written only by the assigned vet; Cancelled is written by staff. The displayed status is derived from this value and appointmentDate

Status rules. Completed is written only when the assigned vet completes the appointment (PATCH /appointments/:id/status), which requires a stored Scheduled appointment whose appointmentDate has passed. The UPDATE is guarded on appointmentStatus = 'Scheduled', so a concurrent cancel or a double submit fails with 409. Optional completion notes insert a HEALTH_RECORD linked to the appointment (appointmentID) in the same transaction. Cancelled is written by staff (PATCH /appointments/:id/cancel).

Displayed status. The displayed status is derived by deriveAppointmentStatus, shared by the Staff, Vet and Adopter APIs: a stored Completed appointment and a past-dated Scheduled appointment both display as Completed. The vet's appointment detail returns both the displayed status and the stored appointmentStatus.

Double-booking guard. A partial unique index on (petID, vetID, appointmentDate) WHERE appointmentStatus = 'Scheduled' guards against a double-submitted booking (see 3.34).

3.20 VOLUNTEER_APPLICATION

Stores applications submitted by prospective volunteers before their account is created.

Column Name	Data Type	Constraints	Description
applicationID	INT	PK, AUTO_INCREMENT	Unique identifier for the application
volunteerID	INT	FK → VOLUNTEER (userID) NULLABLE	Volunteer ID created only if application is approved
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff handling the application
createdAt	DATETIME	NOT NULL	Application creation date and time.
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for shelter volunteer is applying to
applicationStatus	VARCHAR (45)	DEFAULT 'Created' ENUM ('Created', 'Accepted', 'Rejected')	Application status
applicationRemark	VARCHAR (100)	NULLABLE	Any remarks

3.21 VOLUNTEER

Stores account, contact and availability information for volunteers. A volunteer onboards (Personal, Address, government ID, Review) before any staff member at their shelter can approve the account, and accountStatus stays Pending until then. volunteerSchedule holds the volunteer's weekly availability in a compact structured form, described in its column definition.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the volunteer, shared with USERS
volunteerCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "VOL-00005") GENERATED ALWAYS STORED from userID, read-only
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
volunteerName	VARCHAR (45)	NOT NULL	Name of the volunteer
addressLine1	VARCHAR (100)	DEFAULT ''	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ''	City
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the volunteer has finished the onboarding wizard. Volunteers who were not Pending when onboarding was introduced were backfilled TRUE (see 3.35)
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)
volunteerPhone	VARCHAR (20)	NULLABLE	Volunteer contact number
volunteerDOB	DATE	NULLABLE	Date of Birth of volunteer
volunteerSex	CHAR (1)	NULLABLE	Gender of volunteer
volunteerSchedule	VARCHAR (100)	NULLABLE	Weekly availability stored compactly, e.g. "Mon:MA;Wed:E" (days Mon–Sun; M = Morning, A = Afternoon, E = Evening).

Column Name	Data Type	Constraints	Description
			Written by PUT /volunteers/me/availability and decoded to { Mon: ["Morning","Afternoon"], … } in API responses. Older free-text values are left as-is and decode as null.
shelterID	INT	FK → SHELTER (shelterID) NULLABLE	Unique identifier for shelter volunteer works at
createdAt	DATETIME	NOT NULL	Account creation timestamp
accountStatus	VARCHAR (45)	DEFAULT 'Pending' ENUM ('Pending', 'Active', 'Banned', 'Deactivated', 'Closed', 'Declined')	Account status — Pending until a staff member approves (see 3.33)

3.22 GovernmentID

Stores government identity documents for adopters, staff, volunteers, vets and admins for verification purposes.

Column Name	Data Type	Constraints	Description
governmentIDID	INT	PK, AUTO_INCREMENT	Unique identifier for the record.
userID	INT	NOT NULL	The ID of the adopter, staff, volunteer, veterinarian or admin
userType	VARCHAR (45)	NOT NULL ENUM ('Adopter', 'Staff', 'Volunteer', 'Veterinarian', 'Admin')	Designation
idType	VARCHAR (45)	NOT NULL	Type of ID document
idNumber	VARCHAR (45)	NOT NULL	ID value
verificationStatus	VARCHAR (15)	DEFAULT 'Pending' ENUM ('Pending', 'Verified', 'Rejected')	Verification status
documentURL	VARCHAR (500)	NULLABLE	Object path within the private government-ids Storage bucket

3.23 TASKS

Stores volunteer tasks created by staff for a specific shelter branch.

Column Name	Data Type	Constraints	Description
taskID	INT	PK, AUTO_INCREMENT	Unique identifier for the task
taskName	VARCHAR (100)	NOT NULL ENUM ('Animal_Care', 'Vet_Assistance', 'Cleaning', 'Feeding', 'Events', 'Admin', 'Other')	Task Name
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the shelter where task is assigned
taskDesc	VARCHAR (300)	NOT NULL	Task Description
taskDate	DATETIME	NULLABLE	Task creation date
taskDue	DATETIME	NULLABLE	Task due date
taskStatus	VARCHAR (45)	NULLABLE ENUM ('In_progress', 'Completed', 'Cancelled')	Task status
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff responsible for task

3.24 VOLUNTEER_TASK

Junction table linking volunteers to the tasks assigned to them. Staff assign volunteers when a task is created or edited. The FK to VOLUNTEER is ON DELETE RESTRICT (3.34), so a volunteer's rows are removed before the VOLUNTEER row when an account is deleted.

Column Name	Data Type	Constraints	Description
taskID	INT	PK, FK → TASKS (taskID)	Unique identifier for the task
volunteerID	INT	PK, FK → VOLUNTEER (userID)	Unique identifier for the volunteer assigned to task

3.25 EVENTS

Stores adoption events, fundraisers and community programs organised by shelter staff.

Column Name	Data Type	Constraints	Description
eventID	INT	PK, AUTO_INCREMENT	Unique identifier for the event
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the shelter where event takes place.
eventName	VARCHAR (45)	NOT NULL	Name of event

Column Name	Data Type	Constraints	Description
eventCategory	VARCHAR (25)	NOT NULL ENUM ('Adoption_Event', 'Fundraiser', 'Volunteer_Orientation', 'Vaccination_Clinic', 'Community_Outreach', 'Workshop', 'Donation_Drive', 'Other')	Event category
eventDate	DATETIME	NOT NULL	Event date and time
eventDesc	VARCHAR (300)	NOT NULL	Description of event.
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff responsible for managing event

3.26 VOLUNTEER_EVENT

Junction table linking volunteers to the events they are participating in. Staff assign volunteers to events; a volunteer cannot sign themselves up. Rows for events that have not happened yet are removed when a volunteer closes their account, and all of a volunteer's rows are removed when the account is deleted. The FK to VOLUNTEER is ON DELETE RESTRICT (3.34).

Column Name	Data Type	Constraints	Description
eventID	INT	PK, FK → EVENTS (eventID)	Unique identifier for the event
volunteerID	INT	PK, FK → VOLUNTEER (userID)	Unique identifier for the volunteer assigned to event

3.27 DONOR

Stores account and contact information for users who make monetary donations to shelter branches.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the donor, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
donorName	VARCHAR (45)	NOT NULL	Name of donor
addressLine1	VARCHAR (100)	DEFAULT ''	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2

Column Name	Data Type	Constraints	Description
city	VARCHAR (45)	DEFAULT ''	City
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the donor has finished the onboarding wizard
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)
donorPhone	VARCHAR (20)	NULLABLE	Donor contact number
donorDOB	DATE	NULLABLE	Date of Birth of donor
donorSex	CHAR (1)	NULLABLE	Gender of donor
createdAt	DATETIME	NOT NULL	Account creation timestamp
stripeCustomerID	VARCHAR (255)	NULLABLE	Stripe payment ID
accountStatus	VARCHAR (45)	DEFAULT 'Active' ENUM ('Active', 'Banned', 'Deactivated', 'Closed')	Account status (see 3.33)

3.28 DONATION

Stores individual donation transactions made by donors to specific shelter branches. A row is created only by the Stripe checkout.session.completed webhook (metadata.kind = "donation") after payment; the API never inserts one at checkout. donationAmt is the amount Stripe actually charged (amount_total / 100), stored as FLOAT; checkout accepts whole US dollars from 1 to 10,000. Rows survive their donor: donorID is set to NULL when the donor deletes their account (shown to staff as "Former donor"), the same treatment as APPOINTMENTS.vetID.

Column Name	Data Type	Constraints	Description
donationID	INT	PK, AUTO_INCREMENT	Unique identifier for the donation
donationCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "DON-00012") GENERATED ALWAYS STORED from donationID, read-only
donorID	INT	FK → DONOR (userID) NULLABLE	Donor who made the donation. NULL once the

Column Name	Data Type	Constraints	Description
		ON DELETE SET NULL ON UPDATE CASCADE	donor has deleted their account (displayed as "Former donor")
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the shelter
donationDate	DATETIME	NOT NULL DEFAULT now()	Donation date and time
donationAmt	FLOAT	NOT NULL	Amount donated in US dollars: the amount Stripe actually charged (amount_total / 100). Whole dollars 1–10,000 at checkout
donationDesc	VARCHAR (300)	NULLABLE	Description of donation
stripeCheckoutSessionID	VARCHAR (255)	NULLABLE UNIQUE (Donation_stripeCheckoutSessionID_key)	Stripe Checkout Session that produced this donation. The unique index stops a replayed webhook recording a donation twice. NULL for donations recorded before Stripe checkout. Never returned by the API
stripePaymentIntentID	VARCHAR (255)	NULLABLE	Stripe PaymentIntent for the payment. NULL for donations recorded before Stripe checkout. Never returned by the API

3.29 ADOPTER_QUIZ

Stores an adopter's answers to the personality quiz. Answers are stored as JSON and validated by the application against the quiz configuration, so the questions can change without a migration. A save always replaces the whole answer set.

Column Name	Data Type	Constraints	Description
adopterID	INT	PK, FK → ADOPTER (userID) ON DELETE CASCADE ON UPDATE CASCADE	Unique identifier for the adopter the answers belong to
answers	JSONB	NOT NULL	An object mapping each question ID to the chosen option code, e.g. { "aloneTime": "workday", "exercise": "oneToTwoH", … }; always the whole set
updatedAt	TIMESTAMP (3)	NOT NULL	Set by the application on every save (no database default)

3.30 ADOPTER_MATCH

Stores the compatibility matcher's cache, so page loads don't call the AI. One row per adopter and shortlisted pet, holding the rule-based score, the optional AI score, the combined score and the AI's one-sentence reason. Rows are disposable and rebuilt when stale. Saving the quiz, or a profile update that changes any field the matcher reads, deletes all of that adopter's rows. The fields the matcher reads are numChildren, currentPets, openToSpecialNeeds, preferredSize, preferredAgeRange, preferredBreedID, housingType, yardAvailable, petExperience and activityLevel.

Column Name	Data Type	Constraints	Description
adopterID	INT	PK (composite), FK → ADOPTER (userID) ON DELETE CASCADE ON UPDATE CASCADE	Unique identifier for the adopter the match belongs to
petID	INT	PK (composite), FK → PET (petID) ON DELETE CASCADE ON UPDATE CASCADE	Unique identifier for the matched pet
ruleScore	INT	NOT NULL	Rule-based score, 0–100
aiScore	INT	NULLABLE	AI personality-fit score, 0–100; null when the AI didn't score the pet (AI not configured, or the AI call failed)
totalScore	INT	NOT NULL	0.6 × ruleScore + 0.4 × aiScore, rounded; equals ruleScore when aiScore is null
reason	VARCHAR (300)	NULLABLE	The AI's one-sentence "why this match" reason; null for rule-only rows
computedAt	TIMESTAMP (3)	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the computation started

Primary key and index. The primary key is the composite (adopterID, petID). Index AdopterMatch_adopterID_totalScore_idx covers (adopterID, totalScore) (see 3.34).

3.31 PET_HEALTH_SUMMARY

Caches one AI-generated health-passport summary per pet ("Summarize with AI"), so the AI is called again only once the passport changes. The cache key is a hash of the prompt and the pet's records, doses and transfers, so any new or edited record, dose or transfer (or a reworded prompt) regenerates it. Overdue-vaccine points are never stored here; the application computes them from the vaccination records on every request.

Column Name	Data Type	Constraints	Description
petID	INT	PK, FK → PET (petID) ON DELETE CASCADE ON UPDATE CASCADE	Unique identifier for the pet the summary belongs to

Column Name	Data Type	Constraints	Description
inputHash	VARCHAR (64)	NOT NULL	SHA-256 hex of the AI's standing prompt rules plus the passport data sent to it
summary	VARCHAR (1500)	NOT NULL	The AI summary text
pointsToReview	JSONB	NOT NULL	The AI's points to review, each { text, sourceType ("healthRecord" | "vaccination" | "transfer"), sourceID }
generatedAt	TIMESTAMP (3)	NOT NULL	When the summary was generated

3.32 ACCOUNT_TOKEN

Holds the single-use emailed links: verify an email address, reset a password, reactivate a Closed account and confirm an email change. Only the SHA-256 hex of each random token is stored; the raw token exists only in the emailed link. Issuing a new link deletes the user's earlier unused tokens of that purpose, so only the newest link of a purpose works. Tokens belong to USERS rather than to a role table, so every role uses the same mechanism.

Column Name	Data Type	Constraints	Description
tokenID	INT	PK, AUTO_INCREMENT	Unique identifier for the token row
userID	INT	NOT NULL, FK → USERS (userID) ON DELETE CASCADE ON UPDATE CASCADE	The account the link belongs to
purpose	ENUM	NOT NULL ENUM ('EmailVerification', 'PasswordReset', 'Reactivation', 'EmailChange')	What the link does (AccountTokenPurpose, below)
tokenHash	VARCHAR (64)	NOT NULL, UNIQUE	SHA-256 hex of the random token; the raw token is never stored
newEmail	VARCHAR (45)	NULLABLE	EmailChange only: the address the link confirms
expiresAt	TIMESTAMP (3)	NOT NULL	When the link stops working
usedAt	TIMESTAMP (3)	NULLABLE	When the link was used; null while it is unused
createdAt	TIMESTAMP (3)	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the link was issued

AccountTokenPurpose. EmailVerification verifies an email address; PasswordReset sets a new password from the forgot-password link; Reactivation reopens a Closed account (3.33); EmailChange confirms a new login email and is the only purpose that uses newEmail. How long each purpose's link lives is set in the application's email config (config/email.js), and expiresAt records the result.

Why only a hash is stored. The raw token exists only inside the emailed link, and the table keeps just its SHA-256 hex. A copy of the table therefore can't be used to recreate a working link. tokenHash is UNIQUE, so a hash identifies exactly one link.

Indexes and cascade. tokenHash is unique (AccountToken_tokenHash_key). AccountToken_userID_purpose_idx on (userID, purpose) covers looking up one user's tokens of one purpose. The FK to USERS is ON DELETE CASCADE ON UPDATE CASCADE, so deleting an account clears its tokens (see 3.34).

3.33 Account Status Values and Transitions

Every role table has an accountStatus column backed by its own enum. Sprint 9 added Closed to all six enums and Declined to the four roles that start Pending and need approval (STAFF, VETERINARIAN, ADMIN and VOLUNTEER). A value means the same thing in every role's enum.

Role table	Enum	Values	Default
ADOPTER	AdopterAccountStatus	Active, Banned, Deactivated, Closed	Active
DONOR	DonorAccountStatus	Active, Banned, Deactivated, Closed	Active
STAFF	StaffAccountStatus	Pending, Active, Deactivated, Closed, Declined	Pending
VETERINARIAN	VetAccountStatus	Pending, Active, Deactivated, Closed, Declined	Pending
ADMIN	AdminAccountStatus	Pending, Active, Deactivated, Closed, Declined	Pending
VOLUNTEER	VolunteerAccountStatus	Pending, Active, Banned, Deactivated, Closed, Declined	Pending

What each value means.

Status	Meaning
Pending	Signed up and waiting for approval (STAFF, VETERINARIAN, ADMIN and VOLUNTEER only).
Active	The account can be used.
Banned	Available for ADOPTER, DONOR and VOLUNTEER only.
Deactivated	Done by an admin or staff member. Only an admin or staff member can

Status	Meaning
	restore it.
Closed	The owner closed their own account. The only status the owner can undo themselves, through an emailed reactivation link; nobody else can change it.
Declined	Turned down at approval (STAFF, VETERINARIAN, ADMIN and VOLUNTEER only). Final: the account can never be approved later.

Why Closed and Declined were split out of Deactivated. Before Sprint 9, an owner closing their own account, an admin or staff member deactivating it, and a sign-up being turned down at approval were all stored as Deactivated, so they couldn't be told apart. A self-service reactivation link must only work for accounts their owner closed. Closed is therefore its own value: it is the only status the reactivation link accepts, Deactivated stays something only an admin or staff member can undo, and Declined is final.

Why the enum change is its own migration. PostgreSQL can't use a new enum value in the same transaction that adds it. Migration account_status_closed_declined therefore only adds the values (ALTER TYPE ... ADD VALUE on each enum), and the data change that uses one of them comes in the next migration, account_tokens (3.35).

One-off relabel of Admin rows. The account_tokens migration relabels the Admin rows whose own deactivation is on record. An admin who closed their own account is the only Deactivated row whose origin is recorded (ADMIN.statusChangedByID equals the admin's own userID), so those rows become Closed and can be reactivated. Every other Deactivated row stays Deactivated, because self-close and admin/staff deactivation can't be told apart for it.

Allowed transitions. A change made by someone other than the owner must follow the transitions below; anything else is refused with a 409. Closing an account and reactivating it are the owner's own actions and are listed in the same tables.

STAFF, VETERINARIAN and ADMIN (Pending, Active, Deactivated, Closed, Declined):

From	To	Done by	Notes
Pending	Active	Approver	Approval at sign-up
Pending	Declined	Approver	Final; the account can never be approved later
Active	Deactivated	Admin or staff member	
Deactivated	Active	Admin or staff member (admin endpoints)	Only they can restore it
Active	Closed	The owner (DELETE /<role>/me, mode deactivate)	
Closed	Pending	The owner, through the emailed reactivation link	Re-approved the usual way; nobody else can change a Closed account

VOLUNTEER (Pending, Active, Banned, Deactivated, Closed, Declined):

From	To	Done by	Notes
Pending	Active	Any staff member at the volunteer's shelter	Approval at sign-up
Pending	Declined	Any staff member at the volunteer's shelter	Final; the account can never be approved later
Active	Deactivated	Admin or staff member	
Active	Banned	Someone other than the owner	
Deactivated, Banned	Active	Someone other than the owner	Restored to Active
Active	Closed	The owner (DELETE /volunteers/me, mode deactivate)	
Closed	Pending	The owner, through the emailed reactivation link	Re-approved the usual way; nobody else can change a Closed account

ADOPTER and DONOR (Active, Banned, Deactivated, Closed; both are Active from sign-up and never Pending or Declined):

From	To	Done by	Notes
Active	Banned	Someone other than the owner	
Active	Deactivated	Admin or staff member	
Banned, Deactivated	Active	Someone other than the owner	Restored to Active
Banned, Deactivated	Each other	Someone other than the owner	Adopter only: Active, Banned and Deactivated can move among themselves
Active	Closed	The owner (DELETE /<role>/me, mode deactivate)	
Closed	Active	The owner, through the emailed reactivation link	Goes straight back to Active; nobody else can change a Closed account

3.34 Constraints & Indexes

Lists the database-level constraints and indexes that sit outside individual column definitions, including partial, expression and spatial indexes.

Table	Constraint	Purpose
ADOPTION_APPLICATION	UNIQUE (adopterID, petID) WHERE applicationStatus IN ('Pending','Accepted')	One active application per adopter + pet
ADOPTION_APPLICATION	UNIQUE (petID) WHERE applicationStatus = 'Accepted'	One accepted application per pet — a pet goes to one applicant
APPOINTMENTS	UNIQUE (petID, vetID, appointmentDate) WHERE appointmentStatus = 'Scheduled'	Blocks double-submitted bookings
GOVERNMENT_ID	UNIQUE (userID, userType)	One ID per user
SPECIES	UNIQUE (LOWER(speciesName))	Case-insensitive unique names
BREED	UNIQUE (speciesID, LOWER(breedName))	Case-insensitive unique within a species
SHELTER	GIST index on shelterLocation	Fast PostGIS proximity search
APPOINTMENTS	FK vetID → VETERINARIAN (userID) ON DELETE SET NULL	Appointment history survives a vet deleting their account
HEALTH_RECORD	FK vetID → VETERINARIAN (userID) ON DELETE SET NULL	Health records survive a vet deleting their account
HEALTH_RECORD	FK appointmentID → APPOINTMENTS (appointmentID) ON DELETE SET NULL	A note outlives the appointment it was written at
HEALTH_RECORD	INDEX HealthRecord_appointmentID_idx (appointmentID)	Fast lookup of the notes written at an appointment
VACCINATION_RECORD	FK administeredBy → VETERINARIAN (userID) ON DELETE SET NULL	Dose history survives a vet deleting their account
VACCINATION_RECORD	FK appointmentID → APPOINTMENTS (appointmentID) ON DELETE SET NULL	A dose outlives the appointment it was given at
DONATION	UNIQUE INDEX Donation_stripeCheckoutSessionID_key (stripeCheckoutSessionID)	A replayed Stripe webhook can't record a donation twice; NULLs (pre-Stripe donations) are allowed
DONATION	FK donorID → DONOR (userID) ON DELETE SET NULL ON UPDATE CASCADE	Donation history survives a donor deleting their account ("Former donor")
VOLUNTEER_TASK	FK volunteerID → VOLUNTEER	A volunteer's task rows

Table	Constraint	Purpose
	(userID) ON DELETE RESTRICT	are removed before the volunteer row
VOLUNTEER_EVENT	FK volunteerID → VOLUNTEER (userID) ON DELETE RESTRICT	A volunteer's event rows are removed before the volunteer row
ADOPTER_QUIZ	FK AdopterQuiz_adopterID_fkey (adopterID) → ADOPTER (userID) ON DELETE CASCADE ON UPDATE CASCADE	A quiz belongs to its adopter and is removed with the adopter
ADOPTER_MATCH	FK AdopterMatch_adopterID_fkey (adopterID) → ADOPTER (userID) ON DELETE CASCADE ON UPDATE CASCADE	Cached matches are disposable and are removed with the adopter
ADOPTER_MATCH	FK AdopterMatch_petID_fkey (petID) → PET (petID) ON DELETE CASCADE ON UPDATE CASCADE	Cached matches are disposable and are removed with the pet
PET_HEALTH_SUMMARY	FK PetHealthSummary_petID_fkey (petID) → PET (petID) ON DELETE CASCADE ON UPDATE CASCADE	The cached summary is removed with the pet
ADOPTER_MATCH	INDEX AdopterMatch_adopterID_totalScore_idx (adopterID, totalScore)	Fast lookup of an adopter's matches by score
ACCOUNT_TOKEN	UNIQUE AccountToken_tokenHash_key (tokenHash)	A hashed token identifies exactly one link
ACCOUNT_TOKEN	INDEX AccountToken_userID_purpose_idx (userID, purpose)	Fast lookup of one user's tokens of one purpose
ACCOUNT_TOKEN	FK AccountToken_userID_fkey (userID) → USERS (userID) ON DELETE CASCADE ON UPDATE CASCADE	Tokens belong to the account and are removed with it

Note: Partial/expression indexes and generated columns can't be expressed in the Prisma schema, so they live in migration SQL (the baseline migration builds the full schema). Uniqueness of a VACCINE (vaccineName + manufacturer, case-insensitive) is enforced by a service-level check, not by an index. The DONATION unique index and the donorID ON DELETE change were added by migration 20261003130000_donation_stripe (3.35). The ACCOUNT_TOKEN indexes and cascade foreign key were created by migration account_tokens (3.35).

Cascading deletes. The cascades are deliberate: these tables hold either a cache (ADOPTER_MATCH, PET_HEALTH_SUMMARY) or data owned by the adopter (ADOPTER_QUIZ), so closing an adopter account or deleting a pet clears them instead of being blocked by them. ACCOUNT_TOKEN rows belong to the account in the same way: deleting a USERS row deletes its tokens.

3.35 Migrations

The database is PostgreSQL 15 with PostGIS on Supabase, accessed through Prisma. Schema SQL lives in migration folders and is applied with prisma migrate deploy. prisma migrate dev is never used, because it treats the PostGIS column and the generated reference-code columns as drift. A baseline migration builds the schema, and later migrations add to it.

Migration	Kind	Purpose
baseline	Schema	Builds the full schema, including the PostGIS shelterLocation column and its GIST index, the generated reference-code columns (petCode, applicationCode, appointmentCode, volunteerCode, donationCode), and the partial and expression indexes listed in 3.34
staff_onboarding_backfill	Data	Marks every non-Pending staff member as onboarded (onboardingComplete = TRUE, onboardingStep = 5), so accounts approved without the wizard are never sent into it
vet_onboarding_backfill	Data	Marks every non-Pending vet as onboarded (onboardingComplete = TRUE, onboardingStep = 5), the counterpart of staff_onboarding_backfill, so accounts approved without the wizard are never sent into it
20261003120000_volunteer_onboarding_backfill	Data	Marks every non-Pending volunteer (Active, Banned, Deactivated) as onboarded (onboardingComplete = TRUE, onboardingStep = 5), so existing volunteers are not sent into the onboarding wizard now that volunteers onboard before approval. Pending volunteers are left alone, and onboardingComplete keeps its FALSE default for new sign-ups. Same pattern as staff_onboarding_backfill and vet_onboarding_backfill
20261003130000_donation_stripe	Schema	Adds DONATION.stripeCheckoutSessionID (VARCHAR (255), NULL, unique index Donation_stripeCheckoutSessionID_key) and stripePaymentIntentID (VARCHAR (255), NULL); both are nullable because donations recorded before this release have no Stripe session. Makes DONATION.donorID nullable and changes its FK to DONOR (userID) from RESTRICT to ON DELETE SET NULL ON UPDATE CASCADE, so a donor deleting their account keeps the shelter's donation history
20261004120000_adopter_match_quiz	Schema	Creates ADOPTER_QUIZ and ADOPTER_MATCH, the AdopterMatch_adopterID_totalScore_idx index on (adopterID, totalScore) and their cascade foreign keys (AI compatibility matcher)
20261009120000_pet_updated_at	Schema	Adds PET.updatedAt, so the matcher can tell when a pet changed after its matches were computed

Migration	Kind	Purpose
20261012120000_pet_health_summary	Schema	Creates PET_HEALTH_SUMMARY and its cascade foreign key (health-passport AI summary)
account_status_closed_declined	Schema	Adds Closed to AdopterAccountStatus, DonorAccountStatus, StaffAccountStatus, VetAccountStatus, AdminAccountStatus and VolunteerAccountStatus, and Declined to the Staff, Vet, Admin and Volunteer enums, splitting Deactivated three ways (Closed, Declined, Deactivated). Its own migration because PostgreSQL can't use a new enum value in the transaction that adds it; the data change that uses one comes in account_tokens
account_tokens	Schema, Data	Creates the AccountTokenPurpose enum and ACCOUNT_TOKEN, with the unique index AccountToken_tokenHash_key, the AccountToken_userID_purpose_idx index and the cascade foreign key to USERS. Backfills: sets USERS.emailVerified to TRUE for every existing account, so the soft gate (applying to adopt, donating) doesn't block them; and relabels as Closed the Admin rows that are Deactivated with statusChangedByID equal to their own userID (the admin closed their own account). Every other Deactivated row is left as it is

4. ER-Diagram

ER-Diagram: https://dbdiagram.io/d/PetPals-69bf5ca6fb2db18e3bd4b303
ER diagram notes (Sprint 7): the dbdiagram should be updated to match 3.28 and 3.34.

DONATION: add stripeCheckoutSessionID (unique, nullable) and stripePaymentIntentID (nullable). donorID is now optional, so the DONOR → DONATION relationship (cardinality row 27) is drawn as optional on the DONATION side, with ON DELETE SET NULL ON UPDATE CASCADE.

VOLUNTEER: no new column or table. volunteerSchedule keeps its type (VARCHAR (100)) but now holds compact weekly availability (3.21). VOLUNTEER → VOLUNTEER_TASK and VOLUNTEER → VOLUNTEER_EVENT stay as drawn; their FKs are ON DELETE RESTRICT, and staff, not volunteers, create the event rows.

ER diagram notes (Sprint 8): the dbdiagram should be updated to match 3.8, 3.29–3.31 and 3.34.

ADOPTER_QUIZ: add the new table (3.29) with adopterID (PK, FK → ADOPTER), answers (JSONB) and updatedAt. The ADOPTER → ADOPTER_QUIZ relationship (cardinality row 34) is 1:1 and drawn as optional on the ADOPTER_QUIZ side.

ADOPTER_MATCH: add the new table (3.30) with the composite PK (adopterID, petID), ruleScore, aiScore (nullable), totalScore, reason (nullable) and computedAt, many-to-one to ADOPTER and to PET. Together these draw ADOPTER – PET as M:M (cardinality row 35). Add the index on (adopterID, totalScore).

PET_HEALTH_SUMMARY: add the new table (3.31) with petID (PK, FK → PET), inputHash, summary, pointsToReview (JSONB) and generatedAt. The PET → PET_HEALTH_SUMMARY relationship (cardinality row 36) is 1:1 and drawn as optional on the PET_HEALTH_SUMMARY side.

PET: add the new column updatedAt (TIMESTAMP (3), NOT NULL, DEFAULT CURRENT_TIMESTAMP) (3.8).

All four new relationships (ADOPTER_QUIZ → ADOPTER, ADOPTER_MATCH → ADOPTER, ADOPTER_MATCH → PET and PET_HEALTH_SUMMARY → PET) are ON DELETE CASCADE ON UPDATE CASCADE (3.34).

ER diagram notes (Sprint 9): the dbdiagram should be updated to match 3.1, 3.32, 3.33 and 3.34.

ACCOUNT_TOKEN: add the new table (3.32) with tokenID (PK), userID (FK → USERS), purpose (enum AccountTokenPurpose: EmailVerification, PasswordReset, Reactivation, EmailChange), tokenHash (VARCHAR (64), unique), newEmail (VARCHAR (45), nullable), expiresAt, usedAt (nullable) and createdAt. Add the index on (userID, purpose). The USERS → ACCOUNT_TOKEN relationship (cardinality row 37) is 1:M, so a user has zero or more tokens, and is ON DELETE CASCADE ON UPDATE CASCADE (3.34).

Account status enums: add Closed to AdopterAccountStatus, DonorAccountStatus, StaffAccountStatus, VetAccountStatus, AdminAccountStatus and VolunteerAccountStatus, and Declined to StaffAccountStatus, VetAccountStatus, AdminAccountStatus and VolunteerAccountStatus (3.33). No column is added or changed on the role tables.

USERS: emailVerified (BOOLEAN, DEFAULT FALSE) keeps its type; existing rows were backfilled TRUE (3.35).